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Abstract

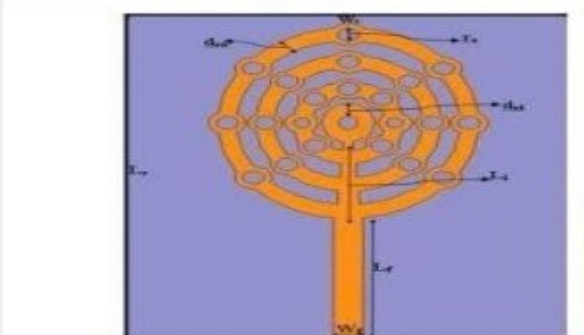
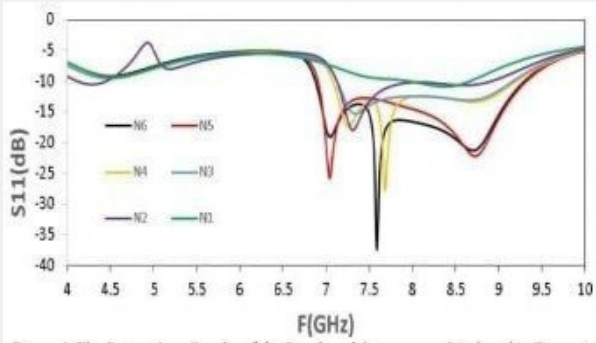
- Next-generation IoT systems require compact antennas with high data rate and reliable communication.
- Quad-port MIMO microstrip patch antennas improve channel capacity and spatial diversity.
- Conventional antenna design involves repeated simulations and high computational time.
- This project designs a quad-port MIMO microstrip patch antenna for sub-6 GHz IoT applications.
- Machine learning techniques are used to predict S-parameters and reduce design complexity.

Keywords: Antenna, S parameter analysis, Mutual coupling Reduction, Next-generation IoT Communication

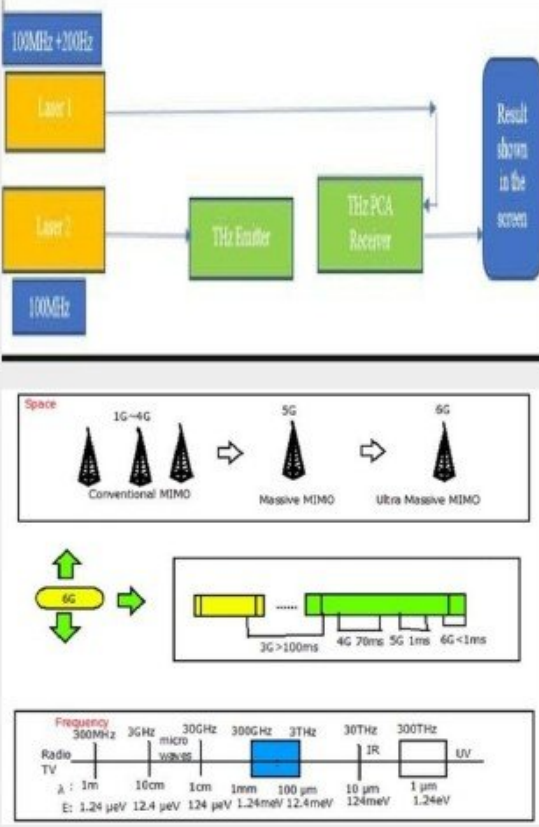
Motivation

- Rapid growth of IoT devices demands compact and efficient antenna systems.
- MIMO antennas are essential to achieve high data rates and reliable communication.
- Designing multi-port antennas using conventional methods is time-consuming.
- Repeated electromagnetic simulations increase design complexity.
- Machine learning offers a faster approach to predicting antenna performance.
- This motivates the use of ML-assisted quad-port MIMO antenna design for IoT applications.

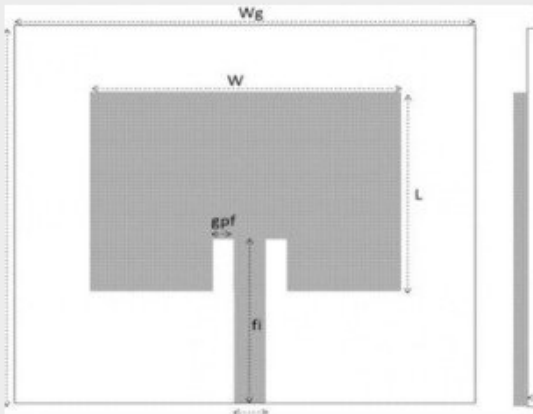
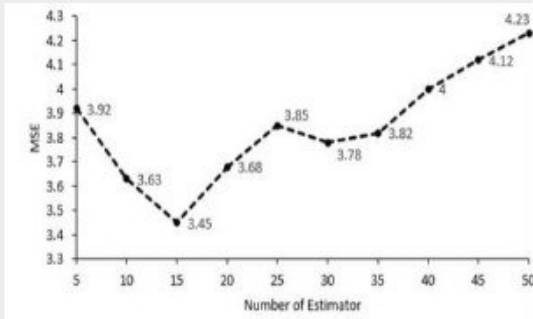
Literature Review

S.NO	Paper Title and Publications details	Structure and results	perceptions	limitations	Material
1.	Microstrip Patch Antennas for Various Applications: A Review Rana et al., Indonesian Journal of Electrical Engineering and Computer Science, 2023	 	Machine learning effectively predicts antenna performance Design time is reduced compared to conventional methods ANN provides better accuracy among the tested models	Dataset is limited to simulation-based values High computational cost for training ML models Experimental validation is not included	Rectangular microstrip patch antenna model Simulation tools and MATLAB for ML implementation

Literature Review

S.No	Paper Tittle And Publication Details	Structure And Result	Perceptions	limitations	Materials
2.	Khan et al., "A Review on Machine Learning and Deep Learning for Various Antenna Design Applications," Heliyon, 2022.	 The diagram illustrates the structure and results of the literature review. It shows a neural network architecture for antenna design, starting with a 100MHz + 200Hz input, followed by Layer 1 and Layer 2, leading to a THz Emitter and a THz PCA Receiver, and finally a Result shown in the screen. Below this, a diagram shows the evolution of MIMO systems from 1G to 6G, including Conventional MIMO, Massive MIMO, and Ultra Massive MIMO. Another diagram shows the frequency spectrum from 300MHz to 300THz, with corresponding wavelengths and energies.	ML significantly reduces antenna design time Neural networks effectively predict resonant frequency and bandwidth Deep learning improves prediction accuracy for complex antenna structure s	Limitation Focused mainly on single-port and simple antenna geometries MIMO antenna configurations are not deeply explored	Material Microstrip and planar antennas MATLAB-based ML and DL models

Literature Review

S.No	Paper Tiltle And Publication Details	Structure And Result	Perceptions	limitations	Materials
3.	Kurniawati et al., "Predicting Rectangular Patch Microstrip Antenna Dimensions Using Machine Learning," Journal of Communications, 2021.	  Fig 4. Random forest MSE	ML provides accurate dimension prediction Design process is faster than conventional methods Regression-based models show good performance	Limited to single-Rectangular element microstrip patch antenna Mutual coupling tools and multi-port effects are not considered	EM simulation and MATLAB

Research Gaps

- Existing ML-based antenna studies mainly focus on single-element patch antennas.
- Limited work is available on applying machine learning to multi-port MIMO antenna systems.
- Mutual coupling effects in quad-port MIMO antennas are not sufficiently addressed using ML.
- Prediction of multi-port S-parameters using machine learning is still limited.

Problem Statement

- Conventional multi-port MIMO antenna design requires repeated simulations, increasing design time and complexity.
- Existing Machine Learning approaches focus mainly on single antennas and do not handle mutual coupling in quad-port MIMO systems.
- Hence, an efficient ML-based approach is needed to predict performance parameters and reduce simulation effort.

Objectives

- To design a quad-port MIMO microstrip patch antenna for IoT applications.
- To analyze S-parameters and mutual coupling between MIMO antenna elements.
- To generate a simulation-based dataset by varying antenna parameters.
- To apply machine learning models for predicting MIMO antenna performance.
- To compare the performance of ML-based prediction with conventional simulation results.

Existing Design

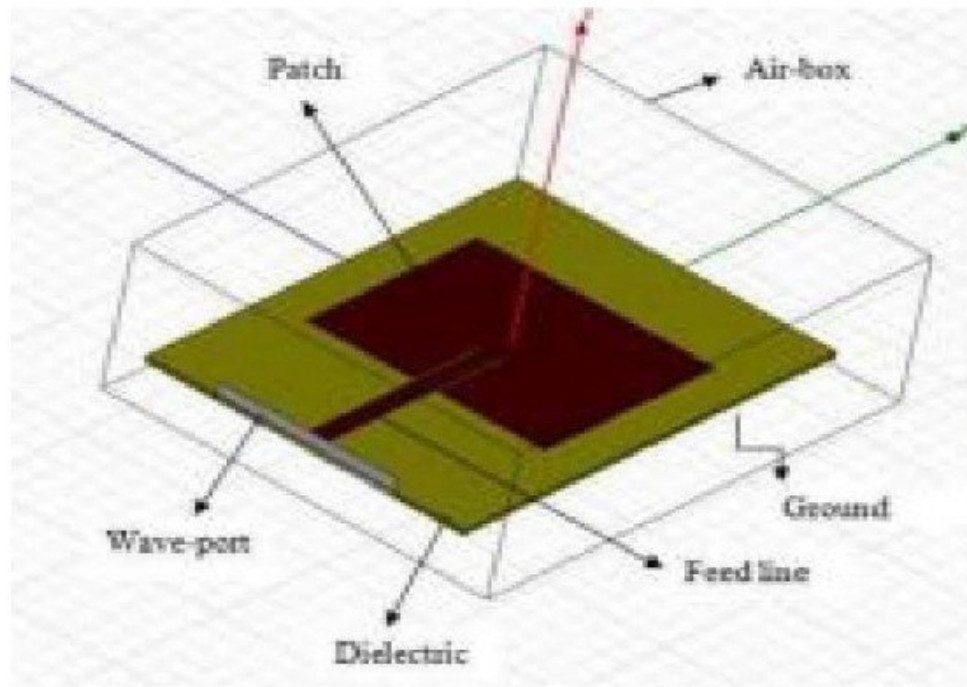


Fig. 1: Single Port MSP Design

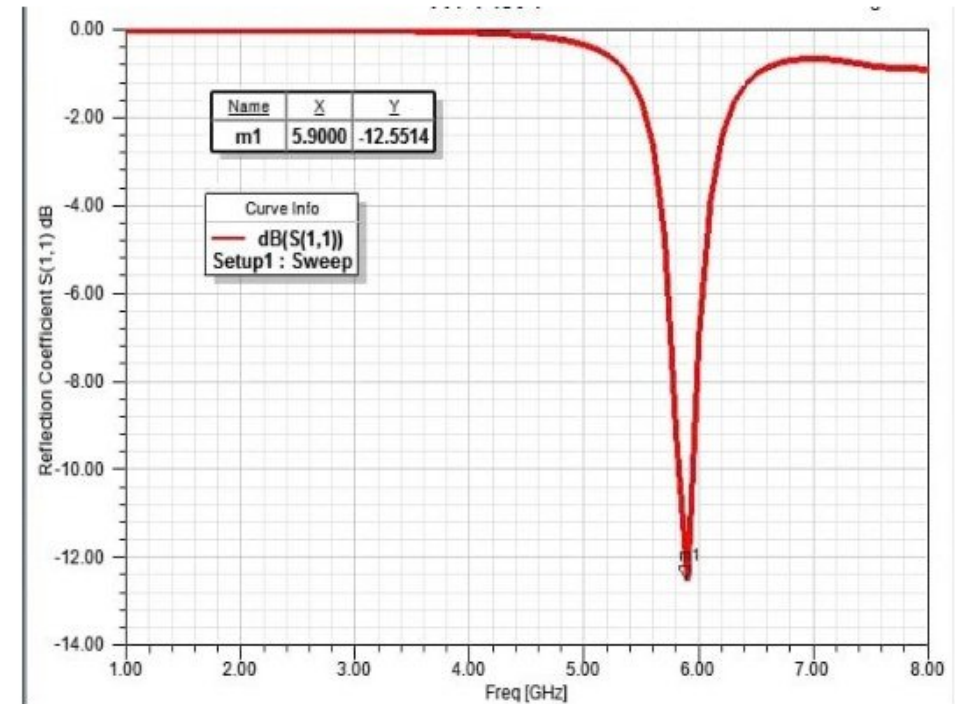


Fig. 2: S parameter of a single Element

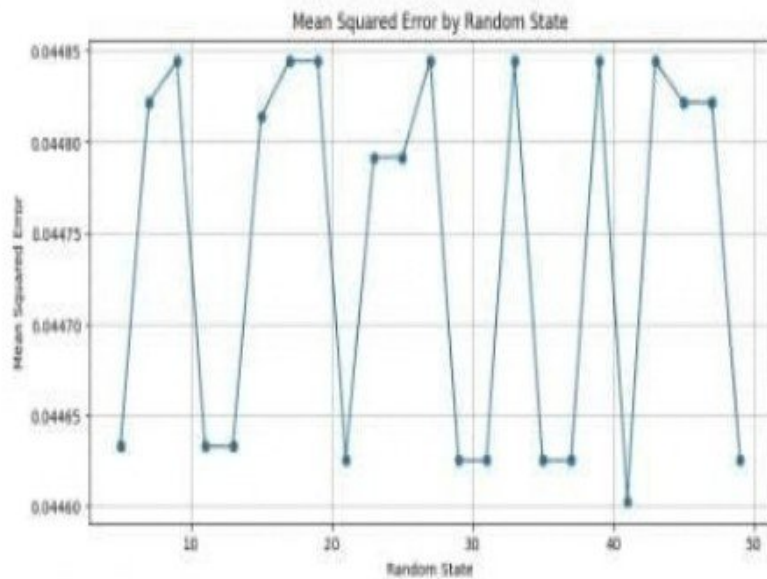


Fig.3: Decision Tree for MSE

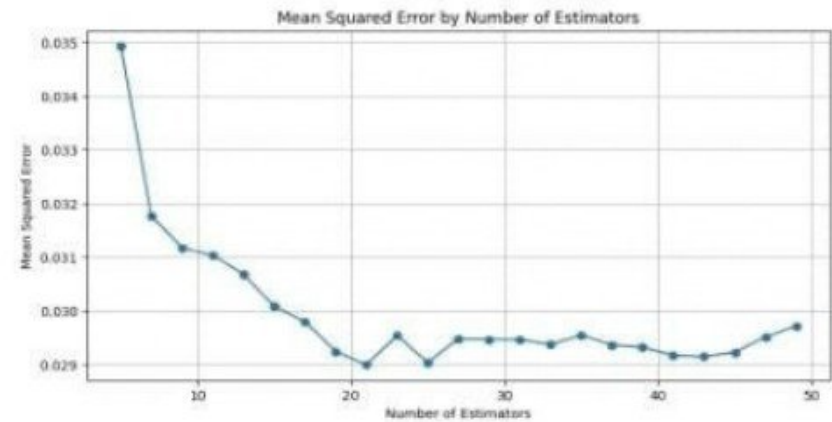


Fig.4: Random Forest for MSE

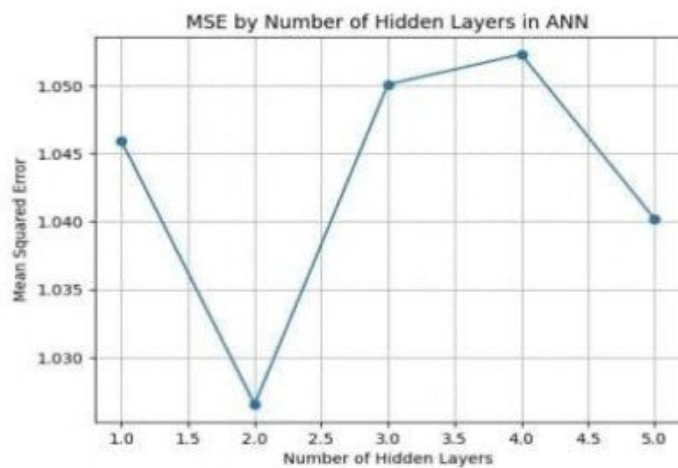


Fig.5: MSE by no.of Hidden Layers in ANN

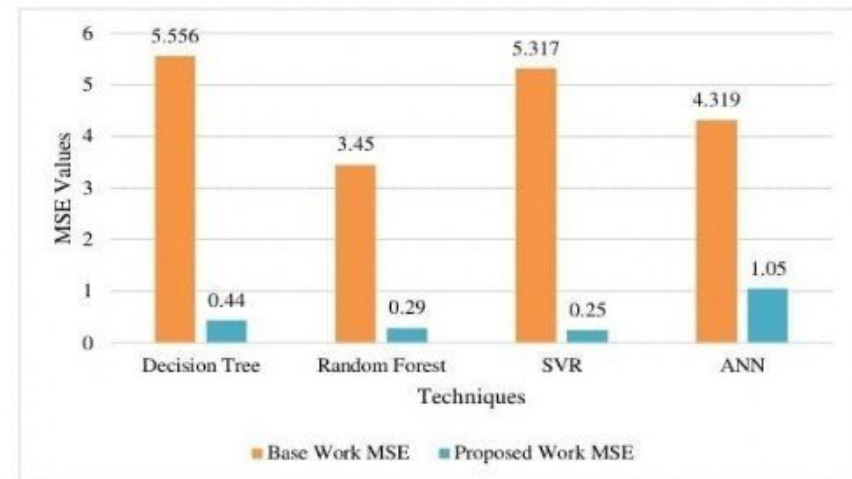


Fig.6: Comparative analysis of MSE values

Proposed Design

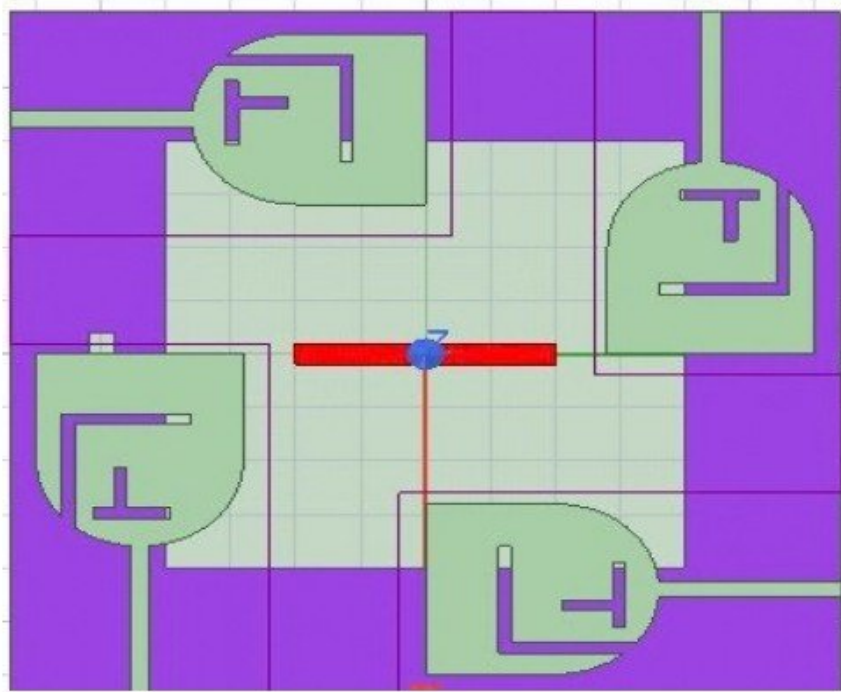


Fig.: Proposed quad-port antenna design without Isolation in top view

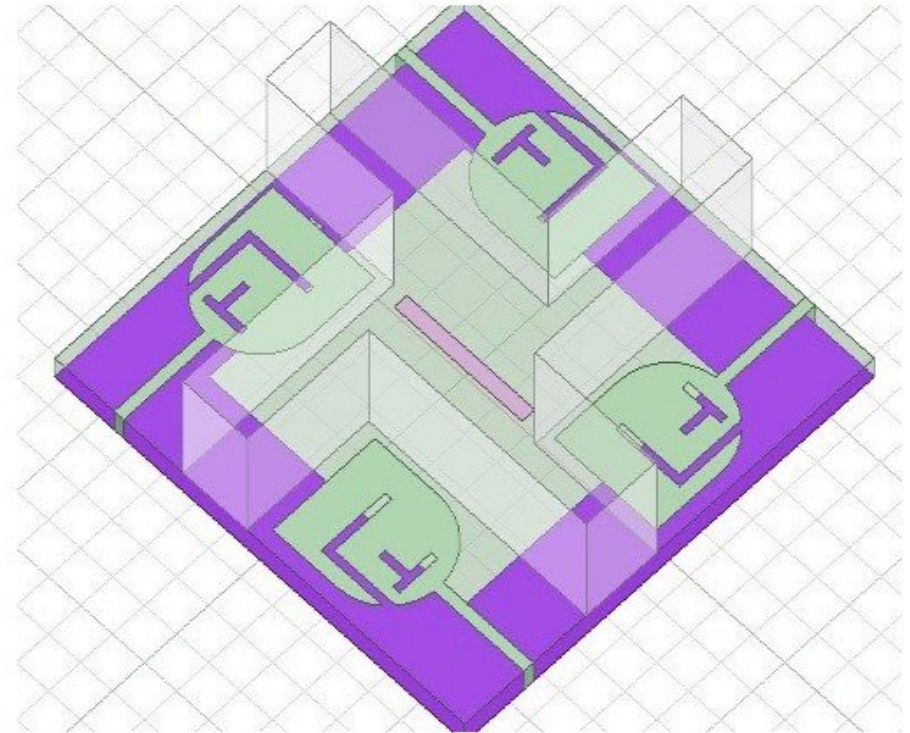


Fig.: Proposed quad-port antenna design with isolation in isometric view

Antenna Design Results

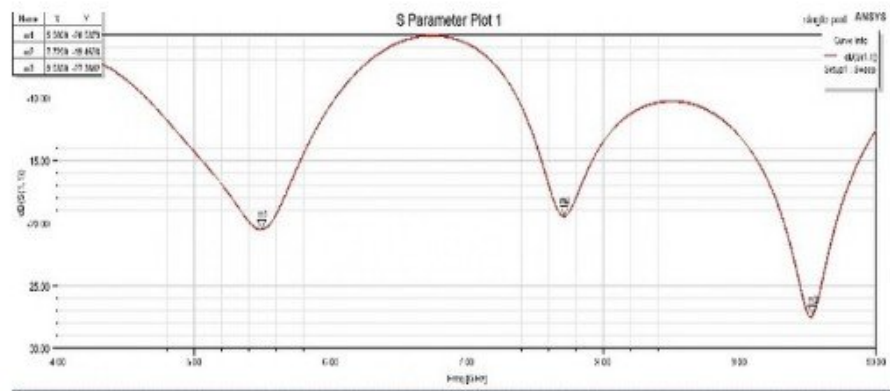


Fig.1:S parameter for single port (Multiband)

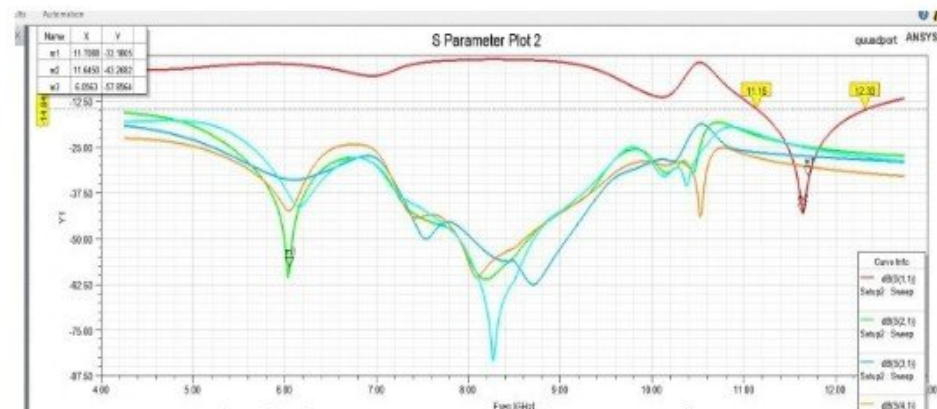


Fig.2:S parameter for quad port (Multiband)

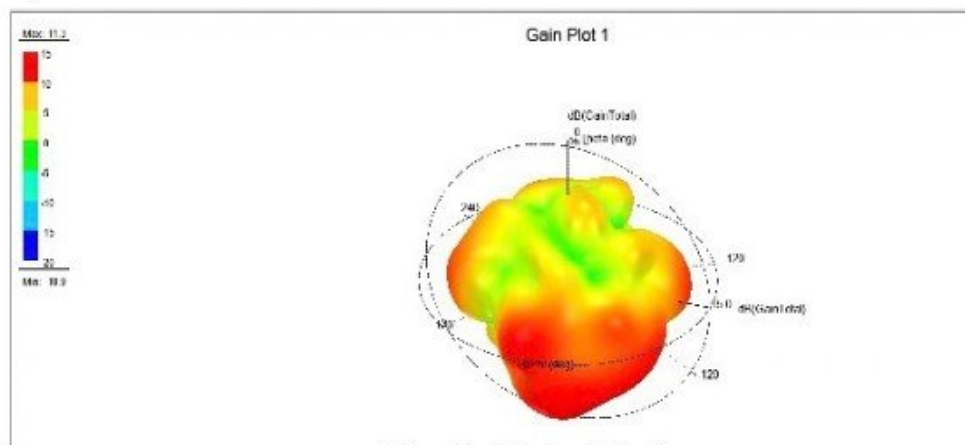
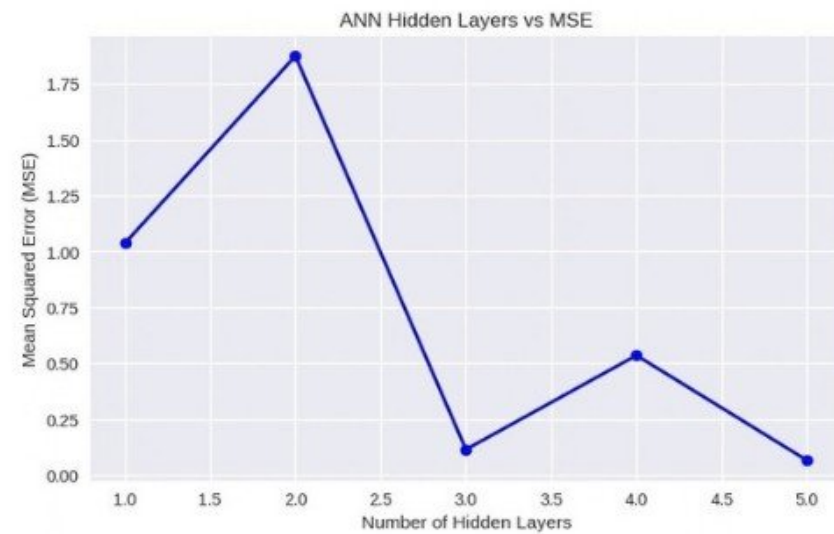
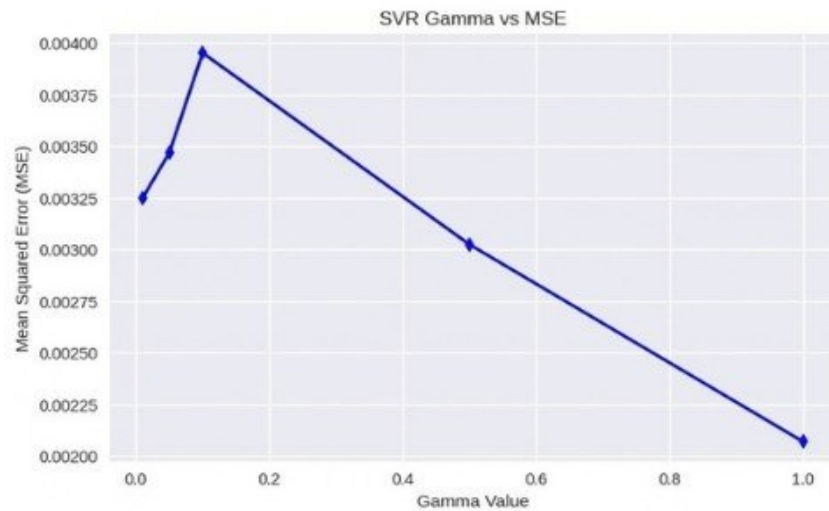
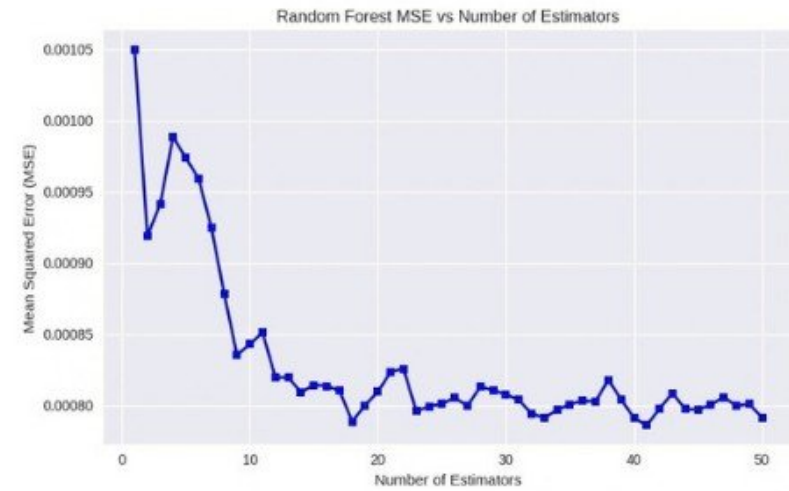
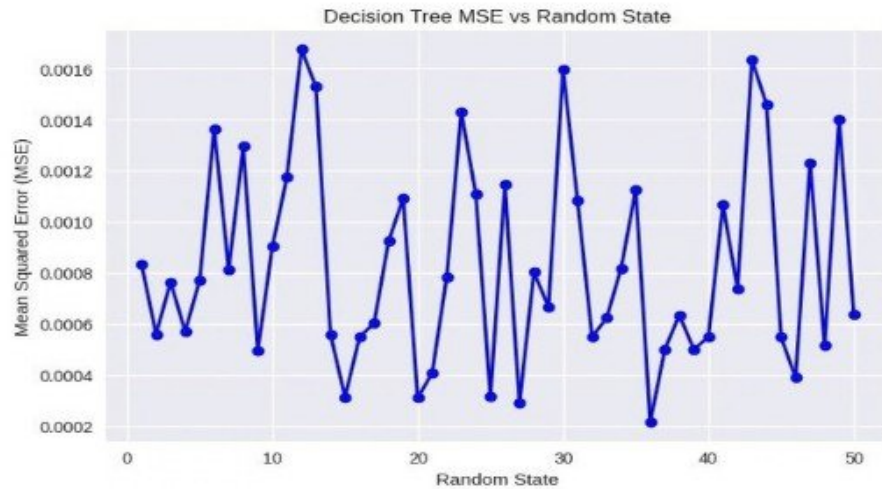
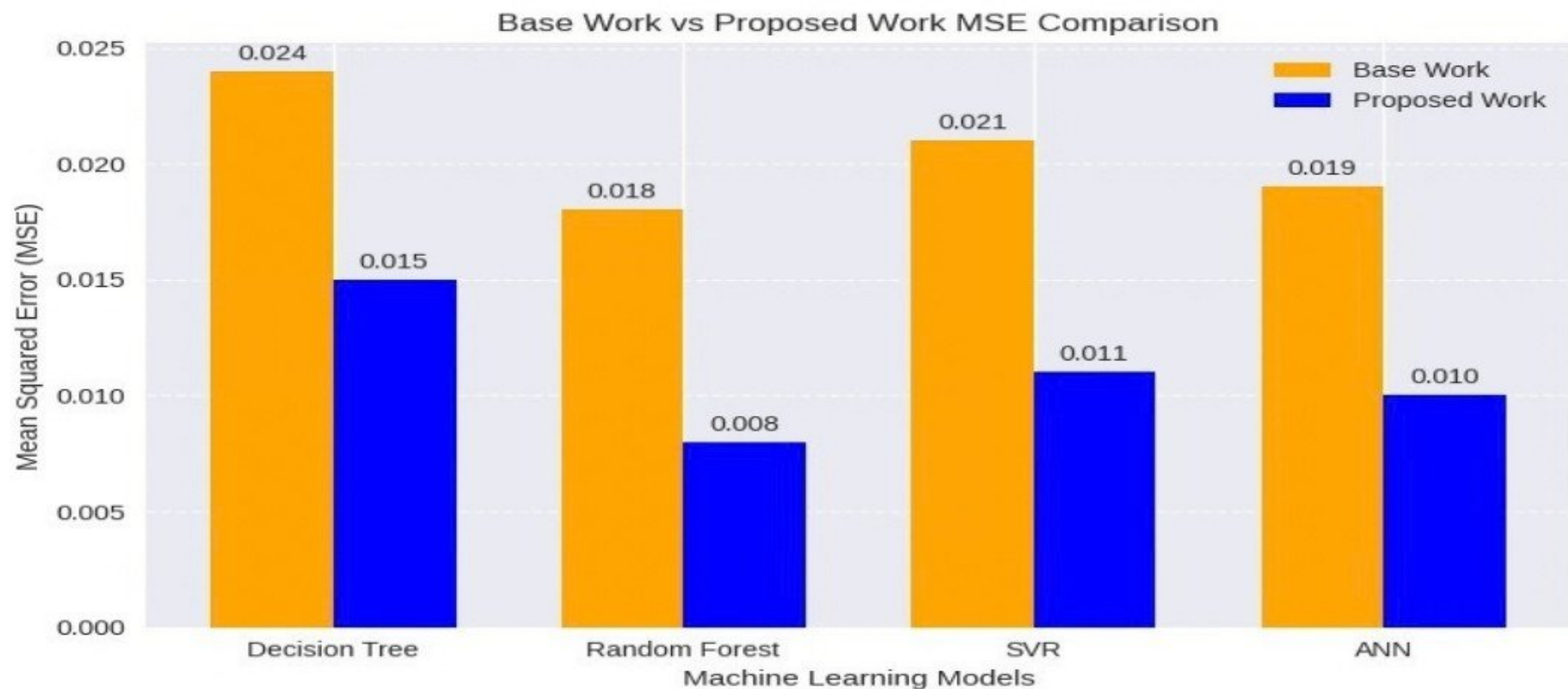


Fig.3: Gain Plot

ML Results



Comparison of existing work& Proposed work



Comparison Table

Model	C o n d i t i o n	MSE	Gai n(d B I)	ECC	DG(dB)	TARC(dB)	I s o l a t i o n (d B)
Decision Tree	Random State=50	0. 00074	6.3	0. 015	9.88	- 12.0	-20
Random Forest	E s t i m a t o r s = 15	0. 00070	6.7	0. 010	9.90	- 12.3	-25
SVR	RBF Kernel	0. 00215	6.4	0. 012	9.85	- 11.8	-22
ANN	Hidden Layers=3	0. 00105	6.5	0. 011	9.87	- 12.0	-23

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- M. S. Rana et al., "Microstrip Patch Antennas for Various Applications: A Review," Indonesian J. of Electrical Engineering and Computer Science, 2023.
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- N. Kurniawati et al., "Predicting Rectangular Patch Microstrip Antenna Dimensions Using Machine Learning," Journal of Communications, 2021.

Thank You...